

Mathematical Toolkit for Grade 11 Physics

Intuitive Calculus, Deductive Derivatives & The Unit Circle Celestial Clock

Target: IELTS Band 6.5+

Curriculum: Vietnam GDPT 2018

Topic: Simple Harmonic Motion

LEARNING OBJECTIVE

Rather than relying on mechanical rote memorization or arbitrary algebraic recipes, this toolkit guides you through the fundamental mathematical machinery that physicists utilize to describe motion: from the **instantaneous rate of change** (the derivative) to the geometric elegance of trigonometric vectors orbiting on the unit circle.

1. What is a Derivative, Really? (The Rate of Change Over Time)

From Average Speed to Instantaneous Speed

Imagine riding a motorbike along a straight road. You cover a spatial distance Δs within an elapsed time interval Δt . The standard parameter representing your overall motion over that entire interval is known as the **average speed**:

$$v_{\text{avg}} = \frac{\Delta s}{\Delta t}$$

However, consider what happens when you abruptly twist the throttle to accelerate and pass a vehicle. If you cast a rapid glance at your speedometer needle at that precise second, what value does the dial actually measure?

That reading represents your **instantaneous speed**—your rate of motion during one infinitesimally brief flash of time. To capture this concept mathematically, physicists conceptually shrink the measurement interval Δt until it approaches zero ($\Delta t \rightarrow 0$). Under this limit, the ratio of distance to time converges to a definite, well-defined value, termed the **derivative of position with respect to time**:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = s'(t) = \frac{ds}{dt}$$

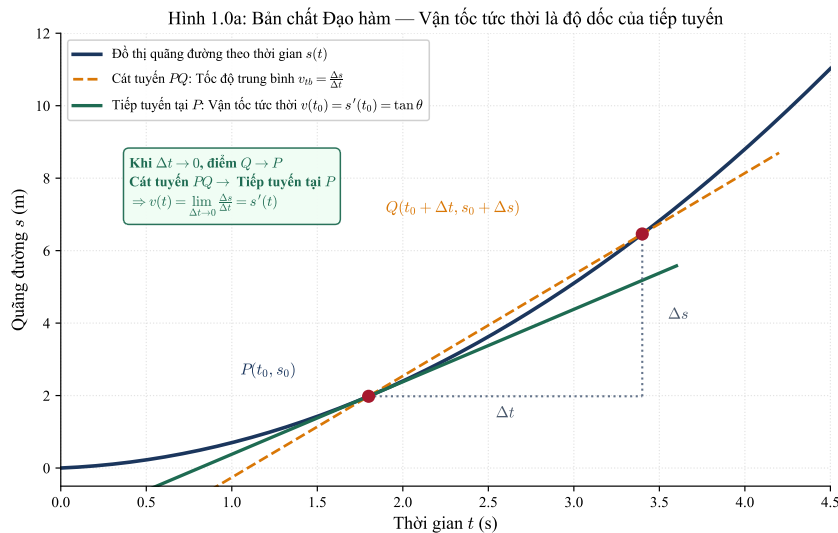


Figure 1.0a: Geometric foundation of the derivative. The secant line PQ illustrates average speed. As $\Delta t \rightarrow 0$, Q glides toward P , morphing into the tangent line whose slope $\tan \theta = s'(t_0)$ defines instantaneous velocity.

- **Geometric Interpretation:** On a distance-time graph (Figure 1.0a), a straight line intersecting two separate coordinates P and Q is called a **secant line**, representing average speed over that span. When point Q slides infinitesimally close to point P ($\Delta t \rightarrow 0$), this secant line pivots smoothly into the **tangent line** resting against the curve at point P .
 - Therefore, **instantaneous velocity is geometrically equivalent to the slope ($\tan \theta$) of the tangent line** touching the position-time curve at that specific instant.
 - A steep upward tangent slope indicates that the object is traveling with high speed.
 - A flat, horizontal tangent line indicates zero slope, meaning the object has momentarily come to rest ($v = 0$).

Acceleration (a) – The Rate of Change of Velocity

In the exact same manner that spatial position changes over time, an object's velocity is rarely static; it increases during acceleration and drops during braking. The quantity that tracks how rapidly velocity transforms across time is called **acceleration**:

$$a(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = v'(t) = s''(t) = \frac{dv}{dt}$$

In concise terms: **Acceleration is the first derivative of velocity**, which simultaneously establishes it as the second derivative of spatial position with respect to time.

International Physics Notation Guide

The alphanumeric symbols featured in secondary school textbooks originate from historic Latin roots and classical English scientific terminology:

Symbol	Scientific Meaning	Etymological Origin
t	Time	English: Time
s	Distance / Arc length	Latin: Spatium (Space / Distance)
x	Displacement coordinate	English: Coordinate (along the horizontal x -axis)
v	Velocity	Latin: Velocitas / English: Velocity (Directed speed)
a	Acceleration	Latin: Acceleratio / English: Acceleration
m	Mass	Latin: Massa / English: Mass
F	Force	Latin: Fortis (Strength) / English: Force
A	Amplitude	Latin: Amplitudo (Maximum displacement from equilibrium)
ω	Angular frequency	Greek alphabet: Omega (Rotational scanning speed of phase)
T	Period of oscillation	English: Time Period (Duration of one complete cycle)
f	Frequency	English: Frequency (Number of oscillations per second)
W	Mechanical Energy / Work	English: Work / German: Werk (W_d : Kinetic, W_t : Potential, W : Total)

2. A Roller Coaster on a Cosine Wave: Uncovering Trigonometric Derivatives & the ω Factor

Transition from Section 1: Formulating the Velocity Problem

In **Section 1**, we established an indispensable physical principle:

“Instantaneous velocity is the derivative of displacement: $v(t) = x'(t)$, which geometrically corresponds to the slope of the tangent line on the position-time curve!”

In simple harmonic motion, an oscillating particle's displacement obeys a pure cosine waveform over time:

$$x(t) = \cos(t)$$

This immediately presents an intriguing deductive problem: **What exact analytical expression represents the velocity function $v(t) = [\cos(t)]'$?**

Rather than blindly memorizing derivative formulas from a table, let us imagine boarding a roller-coaster carriage that glides smoothly across the undulating track $x(t) = \cos(t)$, carefully observing our tangent slope across four distinct checkpoints (**Figure 1.0b(a)**).

A 4-State Deductive Journey: Revealing the Elusive Negative Sign

State 1: The Highest Crest

UNKNOWN SIGN (+sin or -sin?)

- **Parameters:** Time $t = 0$, Displacement $x = +1$ (Positive Amplitude).
- The roller-coaster carriage summits the very highest peak of the hill. At this instantaneous milestone, the carriage ceases its upward ascent and pauses before tipping downward. The track at the summit is completely level and horizontal $\Rightarrow$ **Tangent Slope = 0**.
- Therefore: $x'(0) = 0$ (velocity at the extreme turning point is strictly zero).
- Looking back at our repertoire of introductory trigonometric functions, which standard function evaluates to zero at $t = 0$? The sine function, since $\sin(0) = 0$!
- **A Critical Deduction:** Is the derivative candidate $+\sin(t)$ or $-\sin(t)$? Because both $+\sin(0) = 0$ and $-\sin(0) = 0$, **at the very peak of the hill, the sign of the derivative is strictly UNKNOWN!** The sign remains an unsolved riddle until we witness the track's descent.

State 2: Plunging Downward Through Equilibrium – The Eureka Moment

EUREKA: -sin(t) CONFIRMED

- **Parameters:** Time $t = \frac{\pi}{2}$, Displacement $x = 0$ (Equilibrium Point).
- Departing from the summit, the carriage hurtles downward, rapidly shedding altitude. The vehicle's nose points downward toward the earth $\Rightarrow$ **The tangent slope must be decisively NEGATIVE!**
- Exactly at $t = \frac{\pi}{2}$, as the carriage roars through the equilibrium point $x = 0$, the track plunges at an angle of -45° , registering a steep negative slope of -1 .
- Now, let us test our two competing sign hypotheses against this physical reality:
 - **Hypothesis A (Positive Sign):** If $[\cos(t)]' = +\sin(t)$, then at $t = \frac{\pi}{2}$, we would obtain $+\sin(\frac{\pi}{2}) = +1$ (a positive slope). This blatantly contradicts the observed downward plunge!
 - **Hypothesis B (Negative Sign):** If $[\cos(t)]' = -\sin(t)$, then at $t = \frac{\pi}{2}$, we obtain $-\sin(\frac{\pi}{2}) = -1$ (a negative slope). This matches the physical track slope of -1 with flawless precision!

GROUNDBREAKING FINDING: THE NEGATIVE SIGN REVEALED

The rapid downward plunge through the equilibrium position resolves the mystery once and for all: the derivative of the cosine function **must incorporate an intrinsic minus sign**:

$$[\cos(t)]' = -\sin(t)$$

State 3: The Deepest TroughVERIFIED: $v = 0$

- **Parameters:** Time $t = \pi$, Displacement $x = -1$ (Negative Extreme).
- The carriage reaches the lowest valley floor. For an instantaneous fraction of a second, downward descent halts prior to curving upward. The track is once again flat and horizontal $\Rightarrow$ **Slope = 0**.
- **Analytical Verification:** $-\sin(\pi) = 0$. This aligns completely with the physical fact that velocity vanishes at extreme boundaries.

State 4: Rocketing Upward Through EquilibriumVERIFIED: $v = +v_{\text{max}}$

- **Parameters:** Time $t = \frac{3\pi}{2}$, Displacement $x = 0$ (Equilibrium Point).
- Rocketing out of the valley, the carriage shoots upward past the central equilibrium line, tilted skyward at $+45^\circ \Rightarrow$ **Slope achieves its positive maximum: $+1$** .
- **Analytical Verification:** $-\sin(\frac{3\pi}{2}) = -(-1) = +1$. The formula is completely verified across all quadrants!

The “Time-Compression Machine”: Why Does the ω Factor Emerge?

In Section 1, we recognized ω as the **angular frequency** (the rate at which the oscillation’s phase angle advances over time). Let us now examine the generalized oscillation equation: $x(t) = \cos(\omega t)$. Why does its temporal derivative become $-\omega \sin(\omega t)$ rather than merely $-\sin(\omega t)$?

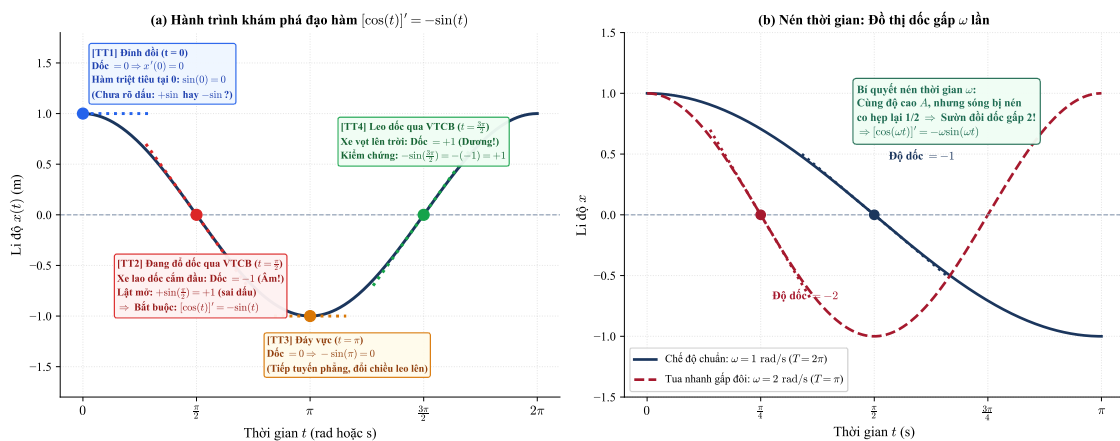


Figure 1.0b: Storytelling trigonometric differentiation: (a) The 4-state roller-coaster expedition revealing the minus sign $[\cos(t)]' = -\sin(t)$; (b) The time-compression effect induced by angular frequency ω , which steepens the curve’s profile by exactly ω times.

Observe the side-by-side comparison in **Figure 1.0b(b)**:

- **Baseline Case ($\omega = 1$):** The wave rolls gently with an extended period $T = 2\pi$. Journeying from the crest ($x = 1$) to equilibrium ($x = 0$) takes an elapsed time of $\Delta t = \frac{\pi}{2}$, generating an equilibrium slope of -1 .
- **Compressed Case ($\omega = 2$):** When the angular frequency doubles, the oscillation’s period is **compressed to exactly half its former duration** ($T = \pi$). The travel window from summit to equilibrium is halved to merely $\Delta t = \frac{\pi}{4}$!
- **The Crucial Geometric Fact:** The height of the summit (amplitude $A = 1$) remains completely unchanged!
- Having to clear the identical vertical drop in only half the available time interval forces the hillside to become **twice as steep**!
- Because the geometric slope of the tangent line is magnified by a factor of ω across every coordinate, the instantaneous rate of change (derivative) at equilibrium surges from -1 to $-\omega = -2$:

$$[\cos(\omega t + \varphi)]' = -\omega \sin(\omega t + \varphi), \quad [\sin(\omega t + \varphi)]' = \omega \cos(\omega t + \varphi)$$

3. Three Satellites on the Celestial Clock: Decoding Phase Angles & the Pythagorean Relation

Connecting Sections 1 & 2: The Kinematic Triad Dilemma

By applying the differentiation mechanics rigorously unlocked in Sections 1 and 2, we establish the fundamental kinematic triad governing simple harmonic motion:

$$\begin{cases} x(t) = A \cos(\omega t + \varphi) \\ v(t) = x'(t) = -\omega A \sin(\omega t + \varphi) \\ a(t) = v'(t) = -\omega^2 A \cos(\omega t + \varphi) \end{cases}$$

At this junction, high school physics students universally confront two intellectual barriers:

1. **The Phase Comparison Barrier:** Position is expressed in $\cos$, velocity is framed in $-\sin$, while acceleration appears as $-\cos$. How can we translate them into a single harmonized $\cos$ format to decisively determine which quantity leads or lags?
2. **The Time-Elimination Barrier:** In practical physics laboratories, detectors record position x and velocity v concurrently at an unknown instant. How can we construct a direct algebraic relationship connecting x and v without relying on a clock to measure time t ?

Standard syllabi typically resolve these hurdles by reciting abstract trigonometric conversion identities:

$$\cos\left(\alpha + \frac{\pi}{2}\right) = -\sin \alpha, \quad \cos(\alpha + \pi) = -\cos \alpha, \quad \cos^2 \alpha + \sin^2 \alpha = 1$$

Instead of straining your cognitive memory with abstract identities, let us examine the **orbital synchronized flight of three satellites navigating a unit circle of radius $R = 1$** (Figure 1.0c):

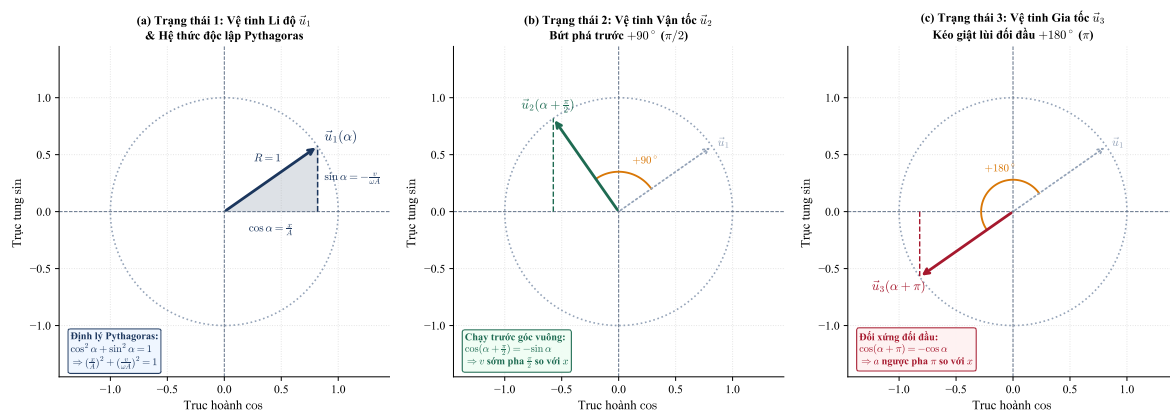


Figure 1.0c: Storytelling 3 satellites orbiting the unit circle: (a) State 1: Displacement satellite $\vec{u}_1$ and the right-angled Pythagorean triangle establishing the time-independent equation; (b) State 2: Velocity satellite $\vec{u}_2$ leading ahead by $+90^\circ (\frac{\pi}{2})$; (c) State 3: Acceleration satellite $\vec{u}_3$ in diametric opposition by $+180^\circ (\pi)$.

The 3-State Satellite Choreography on the Unit Circle

State 1: The Displacement Satellite & The Pythagorean Symphony

TIME-INDEPENDENT EQUATION

Consider a pointing vector $\vec{u}_1$ of constant unit length ($R = 1$), angled at an instantaneous phase orientation $\alpha = \omega t + \varphi$.

- **Horizontal Projection:** Casting the shadow of $\vec{u}_1$ down onto the horizontal axis yields the **normalized displacement**: $\cos \alpha = \frac{x}{A}$.
- **Vertical Projection:** Casting the shadow of $\vec{u}_1$ across onto the vertical axis yields the **normalized velocity magnitude**: $\sin \alpha = -\frac{v}{\omega A}$.
- These two perpendicular geometric shadows, combined with the hypotenuse vector $\vec{u}_1$, construct a **perfect right-angled triangle** (shaded in blue in Figure 1.0c(a))!
- Applying the immortal **Pythagorean Theorem** (adjacent² + opposite² = hypotenuse²):

$$\cos^2 \alpha + \sin^2 \alpha = 1 \Rightarrow \left(\frac{x}{A}\right)^2 + \left(-\frac{v}{\omega A}\right)^2 = 1 \Leftrightarrow \left(\frac{x}{A}\right)^2 + \left(\frac{v}{\omega A}\right)^2 = 1$$

PHYSICAL INSIGHT

The revered “time-independent equation” taught across secondary physics is fundamentally nothing other than **Pythagoras’ Theorem mirrored directly onto the kinematic phase plane!**

State 2: The Velocity Satellite Sprinting +90° Ahead

PHASE LEAD: + $\pi/2$ RADIANS

Velocity is an anticipatory physical quantity that forecasts where an object’s displacement will wander in the immediate future. Consequently, the velocity satellite $\vec{u}_2$ perpetually orbits ahead of the displacement satellite $\vec{u}_1$ by exactly one quarter turn (+90° or + $\frac{\pi}{2}$ radians):

- When rotated +90° counterclockwise into the second quadrant, the original vertical altitude ($\sin \alpha$) tips over horizontally, landing squarely on the negative side of the horizontal axis.
- Therefore, the horizontal abscissa of $\vec{u}_2$ is evaluated as $-\sin \alpha$:

$$\cos\left(\alpha + \frac{\pi}{2}\right) = -\sin \alpha$$

- Substituting this fundamental geometric identity into our dynamic velocity equation:

$$v(t) = -\omega A \sin(\omega t + \varphi) = \omega A \cos\left(\omega t + \varphi + \frac{\pi}{2}\right)$$

PHYSICAL INSIGHT

Velocity v permanently leads displacement x by a phase angle of $\frac{\pi}{2}$ radians (90°)! When displacement has only just arrived at the equilibrium crossing ($x = 0$), velocity has already claimed its maximal summit ($v = v_{\max}$).

State 3: The Acceleration Satellite in Diametric Opposition (+180°)

OPPOSITE PHASE: + π RADIANS

Newton’s Second Law of Motion dictates that an object’s acceleration is inextricably aligned with the net restoring force ($F_{\text{net}} = ma$). Whenever an oscillating mass is displaced outward to the right, the stretched medium exerts an inward tug dragging it back to the left. Accordingly, the acceleration satellite $\vec{u}_3$ orbits in direct diametric opposition, positioned a full half-turn (+180° or + π radians) ahead of $\vec{u}_1$:

- A rotation through +180° translates the vector into point-reflection symmetry through the coordinate origin O . Its horizontal projection is inverted into its precise negative counterpart:

$$\cos(\alpha + \pi) = -\cos \alpha$$

- Substituting this geometric relation into our acceleration expression:

$$a(t) = -\omega^2 A \cos(\omega t + \varphi) = \omega^2 A \cos(\omega t + \varphi + \pi) = -\omega^2 x(t)$$

PHYSICAL INSIGHT

Acceleration a is completely out of phase (π radians or 180°) relative to displacement x ! At the very instant displacement peaks at its positive extreme ($x = +A$), acceleration drops to its minimum negative value ($a = -\omega^2 A$) to pull the particle backward toward equilibrium.